

A NEW THEORY OF CREDIT LINES (WITH EVIDENCE)*

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Abstract

We develop and test a dynamic model in which (latent) credit lines serve as a commitment device not to take on more debt, thereby curbing dilution incentives. Offering bundles of debt and credit lines allows lenders to compete more aggressively leading borrowers to extract the maximum surplus. The model explains a number of empirical patterns, such as that the bulk of credit lines are not utilized and that they are bundled with loans. It also generates new predictions, which we validate empirically with a variety of tests, including two based on natural experiments.

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1 Introduction

Households, firms, and sovereigns typically borrow in markets with multiple competing lenders. In such non-exclusive credit markets, borrowers have an incentive to dilute existing debt, leading to excessive leverage and default risk (Bulow and Rogoff (1988), Fama and Miller (1972), and Hart and Moore (1995)) due to what Admati et al. (2018) call the “leverage ratchet effect,” whereby debt today induces more debt tomorrow (Bolton and Jeanne (2007, 2009), Brunnermeier and Oehmke (2013), and DeMarzo and He (2021)). Although contractual tools like covenants and collateral can ostensibly mitigate the problem, they are often weak and can even backfire (Bizer and DeMarzo (1992), Attar et al. (2019b), and Donaldson, Gromb, and Piacentino (2018, 2021, 2025)).

In this paper, we show that a pervasive contract—the credit line—can address the problem. Even if never drawn, a large credit line can serve as a commitment device not to ratchet up leverage, thereby protecting existing lenders against dilution. The reason is that prospective lenders fear being diluted themselves if the credit line is drawn and, therefore, demand rates so high that a borrower does not take on new loans at all—the option to dilute new debt (via the credit line) deters dilution of old debt (via new debt). We show that a borrower and lender in a non-exclusive market can use a simple contractual device—a debt-credit-line bundle—to replicate the outcome of an exclusive market. As such, lenders’ commitment to lend—to honor the credit line—can make up for the borrower’s lack of commitment not to borrow.

The model is consistent with evidence. It explains why credit lines are (i) bundled with loans (they improve loan terms), (ii) large (they dilute new debt enough to deter it), (iii) often undrawn (they serve as a threat exercised off equilibrium), and (iv) expensive for lenders if drawn (they are not actually drawn in equilibrium). As lenders would prefer not to extend credit lines if they were drawn, the model also makes the new prediction that a decrease in lender commitment should frustrate borrowing and increase leverage. We test that prediction using two natural experiments and find

support for it. We show that the patterns are stronger when dilution protection should matter most, a finding of our theory that is not naturally generated by others.

The environment builds on the literature on dynamic capital structure in which a borrower raises debt dynamically from competing lenders (e.g., Admati et al. (2018), DeMarzo and He (2021)). Debt issuance has benefits, e.g., tax advantages or heterogeneous valuations, and costs, due to, e.g., financial distress. Borrower and lenders incentives, however, are not in general aligned due to non-exclusivity: After borrowing from one lender, the borrower could borrow from another, diluting the first.

We extend this framework by introducing lender competition not only in loan contracts but also in credit lines, namely options to borrow a pre-specified amount at a pre-specified price.¹

We begin with two benchmarks. The first is exclusive competition, in which the borrower can commit to trade with only one lender. In this case, the price captures all of the costs and benefits from debt issuance and, thus, the borrower behaves as a monopolist, issuing the debt level that maximizes surplus. The outcome is effectively that of the static trade-off theory (Kraus and Litzenberger (1973)). Credit lines are unnecessary.

The second benchmark is non-exclusive competition in loan contracts but without credit lines. In this case, the borrower, lacking commitment, gradually increases leverage, passing costs onto existing debt holders. That is the leverage ratchet effect (Admati et al. (2018)). As the duration of exclusivity dt shortens, the effect intensifies, and the borrower's surplus converges to that under competitive pricing (DeMarzo and He (2021), Coase (1972)).

We then study our baseline environment with non-exclusivity and date-0 competition over bundles of loan contracts and credit lines. We first analyze the disciplining role of a credit line in place at any $t > 0$, then solve backwards for the equilibrium bundle offered at $t = 0$.

¹The framework can also capture dynamic monopoly in durable goods markets (Coase (1972)). There credit lines correspond to put options offered by the seller.

Our first main result shows how existing credit lines can deter new borrowing. Anticipating that new debt will trigger dilution via the draw down of a large credit line, lenders become less willing to lend. In equilibrium, this feedback can prevent debt issuance altogether, a mechanism we refer to as a “ratchet anti-ratchet effect.”

Our second main result is a characterization of the equilibrium bundling outcome. When lenders compete via loan-credit line bundles at date 0, the borrower selects the bundle that replicates the equilibrium outcome under exclusivity. Each lender offers a bundle consisting of the loan that maximizes the borrower’s lifetime utility paired with a credit line that deters future dilution, i.e. generates borrower commitment, despite remaining undrawn in equilibrium.

The credit line is an entry-deterrent reminiscent of the latent contracts in papers such as Parlour and Rajan (2001) and Attar et al. (2019a, 2019b). But, whereas in that literature lenders can use latent contracts to extract rents from borrowers, here they use credit lines to appeal to them, explaining why borrowers welcome credit lines in practice. Indeed, *ex post*, lenders would prefer not to honor credit lines, explaining why they must be signed by both parties, not just posted by lenders as in the latent contracts literature. It also points to a discontinuity in the limit: We show that an interval dt of exclusivity, no matter how short, is enough to make the borrower-optimal outcome unique, eliminating the lender-optimal one present in one-shot models.

Credit lines here provide dilution protection akin to what covenants and collateral do elsewhere in the literature on non-exclusivity (e.g., Donaldson, Gromb, and Piacentino (2018, 2025)). The mechanism has the advantage that it is self-enforcing and costless in equilibrium. For example, it does not require that existing lenders monitor covenant violations and is thus in full effect even if loans are sold off separately from credit lines, as is typical in the syndicated loan market.

We conclude our theory work with four extensions. The first two establish robustness, showing (i) that our equilibrium characterization remains intact when lenders

compete in bundles, not just loans, for $t > 0$ (Section 5.1), and (ii) that credit lines continue to enforce exclusivity even when financing needs are uncertain, as long as they can also be state-contingent (Section 5.2).

Our other extensions set the stage for our empirical tests. We show (i) that when lender commitment is weaker, i.e. the likelihood of credit line revocation is higher, the borrower issues more debt (Section 5.3). Finally, we consider imperfect competition and show (ii) that the borrower is more likely to rely on credit lines in more competitive markets, in which exclusivity is harder to enforce (Section 5.4).

We then turn to the empirical analysis, drawing on two decades of loan origination data from Dealscan along with firm capital structure data from Capital IQ, Compustat, and CRSP. Because the value of commitment is increasing in the scope for dilution, the model predicts that credit lines are more likely to be used to enforce exclusivity when dilution risk is higher. Proxying for dilution risk using default risk and credit market competitiveness, we find, consistent with the model, that when dilution risk increases, credit lines are (i) more likely to be bundled with loans, (ii) larger, and (iii) deter future borrowing more.

To provide causal evidence that credit lines deter subsequent borrowing, we use two sources of quasi-exogenous variation. First, we use the introduction of the Supplementary Leverage Ratio, which increased the cost of providing credit lines for certain banks. We find that firms exposed to affected lenders are less likely to obtain credit lines and, consistent with the model, are more likely to borrow again shortly thereafter. Second, we use lender-specific negative shocks, following Chodorow-Reich (2014) and Darmouni (2020), to proxy for an increased likelihood of credit line revocation. Following such shocks, borrowing increases, whereas negative shocks to overall lender health reduce borrowing, as in the standard credit supply literature.

We try to distinguish these findings from what would be predicted by other theories, most notably those focusing on credit lines as liquidity insurance. These theories could also suggest that borrowers borrow more after losing access to credit lines, due

to the increased value of precautionary cash holdings, rather than to the loss of a commitment device. We show, however, that firms do not use their new debt to increase cash, which we think points to the relevance of our theory.

This paper contributes to several literatures. It contributes to the work on credit lines (Holmström and Tirole, 1998; Boot et al., 1987; Berkovitch and Greenbaum, 1991; Kashyap et al., 2002; Sufi, 2009; Acharya et al., 2013, 2014; Campello et al., 2011; Ivashina and Scharfstein, 2010), which has studied their role as liquidity insurance, relationship-building devices, and monitored credit facilities. We show something new: Credit lines can be used as commitment devices not to borrow in the future. This function is distinct from liquidity insurance—it operates precisely when lines are *not* drawn—and generates patterns observed in the data that liquidity-based theories do not predict: that credit lines should be bundled with loans, large, more prevalent in competitive markets, and weakening their credibility should increase leverage rather than cash holdings.

Our paper also contributes to the literature on non-exclusive competition in credit markets (Bizer and DeMarzo, 1992; Parlour and Rajan, 2001; Attar et al., 2019a; Bisin and Guaitoli, 2004; Bennardo et al., 2015), as already touched on above. Our findings contribute to corporate finance by showing that the leverage ratchet effect (Admati et al., 2018; DeMarzo and He, 2021) is self-detering once credit lines are part of the contracting space. Finally, it contributes to the literature on dynamic contracting more broadly (DeMarzo and Fishman, 2007; DeMarzo and Sannikov, 2006; Hartman-Glaser et al., 2025) by showing that allowing lenders to compete via menus of contracts may be sufficient to implement commitment allocations.

Layout. Section 2 presents the model; Section 3, the benchmarks; Section 4, the main results; Section 5, the extensions; Section 6, the empirical analysis. Section 7 concludes. Appendices include proofs, data construction, and supplementary figures and tables.

2 Model

There is a single borrower B and infinitely many lenders. Everyone has deep pockets and lives forever, discounting the future at rate ρ .

B has a project that, while active, yields the following flow payoff:

$$v_t dt = y dt + p_t dQ_t - c Q_t dt, \quad (1)$$

where y is the cash flow over $[t, t + dt)$, Q_t is the stock of outstanding debt (with $Q_0 = 0$), dQ_t is the new debt issued over $[t, t + dt)$, p_t is the unit price of debt issued over $[t, t + dt)$. The unit cost of debt is c . This can be interpreted as the after-tax coupon on perpetual debt (consols) and hence we write $c \equiv (1 - \tau)\delta$ for gross coupon $\delta > 0$ and tax rate $\tau \in (0, 1)$. The project is terminated at a random time which arrives with intensity $\lambda(Q_t)$, where λ is a smooth increasing function satisfying $\lambda(Q) \rightarrow \infty$ as $Q \rightarrow \infty$, capturing default.² The probability the project is active at date s is thus $e^{-\int_0^s \lambda(Q_u) du}$, so B's lifetime utility from date t onward can be written as

$$V_t = \int_0^\infty e^{-\int_0^s \rho + \lambda(Q_{t+u}) du} v_{t+s} ds. \quad (2)$$

Lenders' flow payoff from holding a unit of debt is the coupon δdt . The value of the lenders' stream of coupons on a unit of debt is

$$\Gamma_t = \int_0^\infty e^{-\int_0^s \rho + \lambda(Q_{t+u}) du} \delta ds. \quad (3)$$

Gains from trade between B and lenders arise from the wedge between the lender's benefit δ and B's cost of debt c , capturing the tax advantage of coupon payments: $\delta - c \equiv \delta - (1 - \tau)\delta \equiv \delta\tau$. However, since $\lambda' > 0$, debt issuance increases the probability of project termination.

²This can serve as a reduced form for the endogenous default boundary in DeMarzo and He (2021); indeed, the analysis of our benchmark without credit lines (Section 3.2) is analogous to that of their model.

We define $\Psi(Q)$ to be the total surplus $V_t + \Gamma_t Q$ from the constant debt level Q :

$$\Psi(Q) := \int_0^\infty e^{-(\rho+\lambda(Q))s} (y - cQ + \delta Q) ds = \frac{y + \tau\delta Q}{\rho + \lambda(Q)}. \quad (4)$$

We make the following three assumptions on primitives.

Assumption 1 (Well Defined Exclusive Benchmark). *Ψ is smooth and single-peaked.*

This ensures that the exclusive benchmark problem is well behaved, with a unique interior optimum.

Assumption 2 (Bounded Gains from Trade). *There exists a debt level $\bar{Q} > 0$ s.t. for all $Q \geq \bar{Q}$,*

$$\lambda'(Q) \frac{y - cQ}{\rho + \lambda(Q)} > \tau\delta. \quad (5)$$

This ensures that gains from trade vanish at high levels of debt: For high Q , the marginal increase in default risk outweighs the tax benefit of debt.

Assumption 3 (Decreasing Marginal Cost). *$\lambda''(Q) \leq \frac{2\lambda'(Q)^2}{\rho + \lambda(Q)}$.*³

This ensures that B's value function in the benchmark absent credit lines is decreasing and convex in debt, so our model captures those in the literature (DeMarzo and He (2021)).

Financial markets. At each date t , lenders post bundles consisting of a loan contract (p_t, dQ_t) and a credit line $(\tilde{p}, d\tilde{Q})$. In the baseline, lenders are fully committed to contracts accepted by B, but not to others; in Section 5.3, we allow lenders to revoke accepted contracts with some probability.

Solution concept. The solution concept is Markov perfect equilibrium with state variable equal to B's balance sheet, i.e. his debt and credit lines: At each date, the lenders and B act—lenders post contracts and B chooses at most one and, if he

³A concrete example satisfying all three assumptions is $\lambda(Q) = \frac{\rho Q}{1 - Q}$ for $Q \in [0, 1)$, with $\rho = 0.05$, $\delta = 0.1$, $\tau = 0.6$ (so $c = (1 - \tau)\delta = 0.04$), and $y = 0.048$, for which $\arg \max_Q \Psi(Q) = 0.1$.

has a credit line in place, chooses whether to draw it (in full)⁴ or not—such that (i) they maximize their future lifetime payoffs given their beliefs, (ii) their beliefs are consistent, and (iii) B’s balance sheet is a sufficient statistic for the history with respect to the actions.

For a given credit line in place, whether it is drawn or not is a binary variable. That allows us to simplify notation by omitting the explicit dependence on the credit line; we denote the value function without the credit line in place (i.e. after it has been drawn) by V and with it (i.e. before it has been drawn) by $\tilde{V}$.

We restrict attention to non-explosive processes subject to an equilibrium refinement introduced later, which we think is mild, to help us to show uniqueness (Assumption 4).

3 Benchmarks and Preliminaries

We start with two benchmarks without credit lines. The first is the exclusive allocation in which B can commit to borrow from only one lender, as in the classical trade-off theory, or the static monopolist’s problem. The second is the non-exclusive allocation in which B lacks commitment, as in the leverage ratchet effect, or the dynamic monopolist’s problem.

3.1 Benchmark: Full Commitment

We first consider the problem in which B can commit to future debt issuance; in particular, B can commit not to issue debt to other lenders, so debt dilution is not a concern.

At date 0, B chooses a debt level, Q_t for each t , to maximize the expected present

⁴It turns out that partial draw downs are not optimal in equilibrium anyway, a fact that follows shortly from the convexity of the value function. See footnote 6.

value of his payoffs subject to the lenders' participation constraint:

$$\max_{(Q_t)_{t \geq 0}} \int_0^\infty e^{-\int_0^t \rho + \lambda(Q_u) du} (y + p_t dQ_t - c Q_t) dt \quad (6)$$

$$\text{s.t.} \quad p_t \leq \int_0^\infty e^{-\int_0^s \rho + \lambda(Q_{t+u}) du} \delta ds \quad \forall t. \quad (7)$$

It is straightforward to see that the constraint binds. To see the solution, suppose that B issued debt at only one date, date t , and never again. In that case, the integrands above are constants, and the problem becomes to maximize $e^{-\rho t} \Psi(Q_t)$ over Q_t . The solution is well defined (Assumption 1) and equalizes the marginal revenue to the total marginal cost of debt (to both B and the lender):

$$\delta \tau = \lambda'(Q_t) \Psi(Q_t). \quad (8)$$

The problem being stationary, there are no gains from delaying trade, and it is best to issue only once, at $t = 0$:

Proposition 1 (Exclusive Benchmark). *With commitment/exclusive competition, B issues debt only at $t = 0$ with quantity Q^e at price p^e , where Q^e solves equation (8) and*

$$p^e = \frac{\delta}{\rho + \lambda(Q^e)}. \quad (9)$$

Here, the borrower acts as a monopolist, setting marginal revenue equal to marginal cost. Substituting $c = (1 - \tau)\delta$ and p^e from equation (9) into the FOC, we can re-write the first order condition in equation (8) as:

$$p^e = \frac{c + \lambda'(Q^e) \Psi(Q^e)}{\rho + \lambda(Q^e)}. \quad (10)$$

It follows from inspection that the price is above the marginal cost of debt for B. The marginal cost is the increased cost of coupons c plus the increased termination risk $\lambda'(Q^e)$ times the value of continuing the project to B. But λ' is multiplied by the total

surplus Ψ , which includes the value to the lender as well: Via the price, B takes into account the cost of more debt to the lender. Being a monopolist, he keeps quantities below the competitive level, limiting debt issuance to keep the price above marginal cost.

3.2 Benchmark: No Commitment/Non-Exclusivity

We now consider the case in which B can issue debt continuously but cannot commit to his future issuance, a setting that resembles DeMarzo and He (2021). As allocations now need to be time-consistent, we can solve the recursive formulation of the problem. To build intuition, we carry out the key calculations to derive the HJB as the limit of the value function over discrete increments dt :

$$V_t = v_t dt + \frac{1 - \lambda_t dt}{1 + \rho dt} V_{t+dt}. \quad (11)$$

Using that $V_t = V(Q_t)$ by the Markov property and the expression for v_t in equation (1), we have

$$V(Q) = \max_q \left\{ (y + p(Q + qdt)q - cQ) dt + \frac{1 - \lambda(Q)dt}{1 + \rho dt} V(Q + qdt) \right\}, \quad (12)$$

where B takes the price function p as given. In the limit as $dt \rightarrow 0$,

$$(\rho + \lambda(Q))V(Q) = \max_q \left\{ p(Q) + V'(Q) \right\} q + y - cQ, \quad (13)$$

for any smooth policy $qdt = dQ_t$. The objective is linear in the control q , so it must either be zero or have coefficient zero: $q = 0$ or

$$p(Q) + V'(Q) = 0, \quad (14)$$

an equation that also appears in DeMarzo and He (2021).⁵ The solution is characterized in what follows:

Proposition 2 (Non-exclusive Benchmark). *There is a unique smooth equilibrium.*

The value function is

$$V(Q) = \frac{y - cQ}{\rho + \lambda(Q)}, \quad (15)$$

the price is

$$p(Q) = -V'(Q) = \frac{c + \lambda'(Q)V(Q)}{\rho + \lambda(Q)}, \quad (16)$$

and the issuance policy is

$$q = \frac{\tau\delta - \lambda'(Q)V(Q)}{V''(Q)}. \quad (17)$$

Equation (16) says that the price equals (the present value of) B's marginal cost. That is, the price is competitive, per the Coase Conjecture on a durable goods monopolist (Coase (1972)). B would like to ration quantities to keep prices above marginal cost, but is always tempted to issue more. The result also implies that V is convex, a result that is formalized in the following Lemma.

Lemma 1 (Convex Value). *The value function V is decreasing and convex.*

Before moving on, we characterize how the solution changes when buybacks are not allowed, something that we use later on:

Lemma 2 (No Buybacks Solution). *Suppose buybacks are not allowed, i.e. $q \geq 0$. Then, if $Q_t \geq \bar{Q}$, $q \equiv 0$ and V is as in equation (15); otherwise, Proposition 2 holds.*

This result follows from two facts and one line of algebra: If $q \equiv 0$ then (i) V is as in the non-exclusive benchmark (equation (15)) and (ii) p has the same form as in the exclusive benchmark but is evaluated at Q_t (equation (9)). Those facts capture that under non-exclusivity, V is as if B never issues again (hence V) and, under

⁵There is the flavor of mixed-strategy equilibrium here, as the optimum is determined not by B's direct incentives but by the market's need to make him indifferent over his choice of control.

exclusivity, B actually never issues again (hence p). It remains to verify that, for $Q_t \geq \bar{Q}$, $p(Q) + V'(Q)$, the coefficient of q in the HJB is non-positive. Differentiating and rearranging reveals that the condition is exactly Assumption 2, that gains from debt issuance are dissipated if and only if $Q \geq \bar{Q}$.

We assume the equilibrium described in this section (Proposition 2 and Lemma 2) is played whenever credit lines are not available. DeMarzo and He (2021) give conditions for the analogous equilibrium in their environment to be unique, even allowing for policies with jumps.

4 Results

We now study our baseline model, in which B lacks commitment and lenders compete in bundles. We assume here that lenders can offer loans at any date but bundles with credit lines only at date 0, but relax that in Section 5.1.

We show that when lenders can bundle credit lines with their loan offers, the commitment allocations of Section 3.1 obtain. We do so in three steps. First, we show that there is a ratchet effect for credit lines: B draws a credit line if and only if his debt is sufficiently high (Section 4.1). Second, we show that, as a result, credit lines can act as a self-detering mechanism that prevents lenders from making loan offers, which we call the “ratchet-anti-ratchet effect” (Section 4.2). Third, we show that competition on bundles of loans and credit lines implement the commitment allocation (Section 4.3).

4.1 The Ratchet Effect for Credit Lines

Below, we show an analog of Admati et al. (2018)’s leverage ratchet effect, which says that drawing a credit line becomes more attractive as debt increases, a consequence of the convexity of V .

Before that, we define the *associated debt level* for a given credit line $(\tilde{p}, d\tilde{Q})$,

which we denote by $\mathcal{Q}$, as the debt level at which B is indifferent between drawing that credit line and never issuing debt again. Recalling that, by Proposition 2, he gets $V(\mathcal{Q})$ in that case, we have the following definition:

Definition 1 (Associated Debt Level). *For a credit line $(\tilde{p}, d\tilde{Q})$, $\mathcal{Q}$ is an associated debt level if B is indifferent between drawing $(\tilde{p}, d\tilde{Q})$ with debt $\mathcal{Q}$ and never issuing debt again.⁶*

$$\tilde{p}d\tilde{Q} + V(\mathcal{Q} + d\tilde{Q}) = V(\mathcal{Q}). \quad (18)$$

This can be re-written to reflect that the price of the credit line must exactly compensate B for the costs of a higher debt level at $\mathcal{Q}$,

$$\tilde{p} = -\frac{V(\mathcal{Q} + d\tilde{Q}) - V(\mathcal{Q})}{d\tilde{Q}}. \quad (19)$$

We now show that B draws a credit line whenever his debt exceeds its associated debt level:

Proposition 3 (Ratchet Effect). *Consider any equilibrium in which B has a credit line $(\tilde{p}, d\tilde{Q})$ in place with associated debt level $\mathcal{Q} \leq Q^e$ and $d\tilde{Q}$ sufficiently large. Then, B (weakly) prefers to draw the credit line if and only if $Q \geq \mathcal{Q}$.*

Intuitively, because V is decreasing and convex, when B has more debt the cost of issuing at $\tilde{p}$ is relatively low for high Q . Hence B prefers to draw if $Q \geq \mathcal{Q}$, the indifference point, and not to otherwise.

We conclude this section with an assumption on the set of equilibria we study. We think it is mild, as it matters only if B is indifferent between drawing and not.

Assumption 4 (Tie-breaking Rule). *Suppose B has a credit line $(\tilde{p}, d\tilde{Q})$ and debt Q in place. If he is indifferent between drawing and not, he does not draw if and only if Q solves equation (18).*

⁶As foreshadowed in footnote 4, drawing is full or not at all. Drawing down another fraction ϕ would be optimal only if $\tilde{p}\phi d\tilde{Q} + V(Q + \phi d\tilde{Q})$ had a maximum for $0 < \phi < 1$. But that function is convex in ϕ — $(\tilde{p}\phi d\tilde{Q} + V(Q + \phi d\tilde{Q}))'' = V''(Q + \phi d\tilde{Q})d\tilde{Q}^2 > 0$ since $V'' > 0$. So there cannot be an interior maximum; any interior critical point must be a minimum.

The assumption says that B breaks indifference toward not drawing only if he is indifferent between drawing the line (the LHS of (18)) and never borrowing again (the RHS of (18)). The next result says that (i) for any debt level, there is a credit line for which the condition is satisfied and that, given that credit line, (ii) there is no other debt level for which it is.

Lemma 3 (Existence and Uniqueness). *For any debt level Q and size of credit line $d\tilde{Q}$, there is a unique price such that equation (19) holds.*

For any credit line $(\tilde{p}, d\tilde{Q})$, there is at most one debt level for which it does.

Combined with Proposition 3 and Assumption 4, the result says that B draws any credit line if and only if his debt Q exceeds the credit line's associated debt level $\mathcal{Q}$.

4.2 The Ratchet-Anti-Ratchet Effect

Next, we use the result in Proposition 3 to show that for any debt level $\mathcal{Q} \leq Q^e$ subject to no buybacks, there is an associated credit line $(\tilde{p}, d\tilde{Q})$ with $d\tilde{Q}$ large that makes lenders unwilling to post a loan contract that B is willing to accept.⁷ As a result, the debt level $\mathcal{Q}$ is an absorbing state for the process Q_t . The result follows from the ratchet effect in Proposition 3, i.e. that if B takes on any debt he also takes on $d\tilde{Q}$, and Assumption 2, i.e. that lenders' value drops below B's in anticipation of his debt going up enough.

To see why, suppose first that B has a credit line $(\tilde{p}, d\tilde{Q})$ in place with associated debt level $\mathcal{Q}$ (Definition 1) as in Proposition 3. So at $\mathcal{Q}$, B will choose to draw the line if he takes on any new debt $dQ > 0$. Now B prefers *not* to take on the new debt at price p if his payoff from taking the loan and drawing the credit line is less than that of just drawing the credit line:

$$pdQ + \tilde{p}d\tilde{Q} + V(\mathcal{Q} + dQ + d\tilde{Q}) \leq \tilde{p}d\tilde{Q} + V(\mathcal{Q} + d\tilde{Q}). \quad (20)$$

⁷The no buybacks restriction need not be technological, but instead obtains in equilibrium from a lender's commitment not to sell back outstanding debt claims to B, as shown in Proposition 5. It could also reflect financial constraints/shallow pockets, if B does not have cash to do buy backs.

Rearranging gives an upper bound on the price p of new debt:

$$p \leq \frac{V(\mathcal{Q} + d\tilde{Q}) - V(\mathcal{Q} + dQ + d\tilde{Q})}{dQ}. \quad (21)$$

The inequality must hold for all dQ . Thus, given V is convex (Lemma 1), for any p , it is necessary and sufficient that it holds as $dQ \rightarrow 0$. That limiting condition is:

$$p \leq -V'(\mathcal{Q} + d\tilde{Q}). \quad (22)$$

Condition (22) says that B does not want to issue at a price below marginal cost. The twist is that the marginal cost is conditional on having drawn the line. Given B will draw, immediately increasing his debt by $d\tilde{Q}$, it is the marginal cost at $\mathcal{Q} + d\tilde{Q}$, not at just $\mathcal{Q}$, that matters.

We now turn to whether lenders are willing to lend. A lender that anticipates B will draw the credit line is willing to offer a loan (p, dQ) if and only if its value exceeds the price, or

$$\Gamma(\mathcal{Q} + dQ + d\tilde{Q}) \geq p. \quad (23)$$

Comparing inequalities (22) and (23), it follows that for $d\tilde{Q}$ large enough, lenders cannot make an offer that B will accept, as Assumption 2 implies that gains from trade vanish for $\mathcal{Q} + d\tilde{Q} \geq \bar{Q}$ (in particular that $\tau\delta < \lambda'(\mathcal{Q} + d\tilde{Q})V(\mathcal{Q} + d\tilde{Q})$).

Proposition 4 (Ratchet-Anti-ratchet). *For any debt level $\mathcal{Q} \leq Q^e$, there exists an associated credit line $(\tilde{p}, d\tilde{Q})$ with $d\tilde{Q}$ large enough and subject to no buybacks so that*

$$\Gamma(\mathcal{Q} + d\tilde{Q}) < -V'(\mathcal{Q} + d\tilde{Q}). \quad (24)$$

With such credit line in place, the value function is as follows:

$$\tilde{V}(Q) = \begin{cases} \frac{\delta}{\rho + \lambda(\mathcal{Q})}(\mathcal{Q} - Q) + V(\mathcal{Q}) & \text{if } Q \leq \mathcal{Q}, \\ \tilde{p}d\tilde{Q} + V(Q + d\tilde{Q}) & \text{otherwise,} \end{cases} \quad (25)$$

with V as in Proposition 2.

The main takeaway from Proposition 4 is that any $Q \leq Q^e$ can be supported as an absorbing state by a sufficiently large credit line and lender's commitment not to sell back $Q + d\tilde{Q}$.⁸ Moreover, if B's debt is below $\mathcal{Q}$, then B issues debt to jump to the absorbing state $\mathcal{Q}$; when it is above $\mathcal{Q}$, B draws the credit line. At $Q = \mathcal{Q}$, B is indifferent between drawing or not, which given the tie-breaking rule, implies B will not draw (Assumption 4).

There are three key steps to the proof, which is in Appendix B. The first is to show that given a credit line with the properties in the proposition, $\mathcal{Q}$ is an absorbing state, in line with the preceding discussion (shown in proof of Proposition 3). The second is a characterization of the value function given such a credit line is in place. The last is to show existence.

Per footnote 7, the no buybacks condition can be implemented via creditor commitment without any new assumptions on technology. Still, its necessity is an artifact of the continuous time limit, not the economic environment. To see why, suppose time is discrete with period length $dt > 0$. Then, even without any commitment, a gap remains between lenders' value and the borrower's marginal cost for one full period after the credit line is drawn, which is sufficient to deter issuance as in Proposition 4. We therefore view it as a technical assumption.⁹ That is the analog of a point made

⁸Thus the credit line is part of the implementation of an optimal contract as in, e.g., DeMarzo and Fishman (2007), DeMarzo and Sannikov (2006), and Hartman-Glaser et al. (2025). Unlike in that literature, the credit line is latent, consistent with low utilization (Section 6). That also contrasts with models in which credit lines are used for state-contingent liquidity insurance (e.g., Holmstrom and Tirole (1997) and Santos and Viswanathan (2020)).

⁹We can formalize the argument in equations, albeit not fully rigorously. Suppose that the time

heuristically by DeMarzo and He (2021): Away from the continuous-time limit, their firm has a strict incentive to increase leverage. With the credit line, our firm has a strict incentive not to.

4.3 Bundling

We now consider date 0, when lenders can offer bundles including loans and credit lines. The outcome is the exclusive contracting outcome of Proposition 1:

Proposition 5 (Credit Line Bundles). *If lenders compete in bundles, B chooses a bundle with a loan and a credit line at date 0, never borrows again, and never draws the credit line.*

The loan coincides with exclusive outcome (p^e, Q^e) in Proposition 1.

The credit line $(\tilde{p}, d\tilde{Q})$ has associated debt level $\mathcal{Q} = Q^e$ (Definition 1) and is subject to no buybacks.

The equilibrium allocation is unique (under Assumption 4).

Proposition 5 builds on Proposition 4, which says that a credit line is a commitment device not to borrow in the future. At date 0, B can put a credit line in place and thereby commit not to borrow from anyone else. Thus if lenders can bundle loans with credit lines, and commit to such credit lines, then an instant of exclusivity—from date 0 to date dt —is just as good as exclusivity forever. As a result, competition in bundles at date 0 can achieve the same outcome as exclusive competition.

increment dt is bounded away from zero today but that dt^2 is still vanishingly small and that time is continuous forever after. Assume also that dQ is close to zero. Then the condition for B to issue is as in equation (22), but that for lenders to lend becomes

$$p \leq \Gamma(Q + d\tilde{Q}) = \frac{1 - \lambda(Q + d\tilde{Q})dt}{1 + \rho dt} (\delta dt + p_{t+dt}).$$

Substituting $p_{t+dt} = -V'(Q + d\tilde{Q})$ from Proposition 2 and rearranging gives the following condition for there to be no trade, i.e. for the interval of feasible prices to be empty:

$$\delta(1 + \rho dt) < -(\rho + \lambda(Q + d\tilde{Q}))V'(Q + d\tilde{Q}).$$

Substituting for V' from Proposition 2 reduces this to Assumption 2 evaluated at $Q + d\tilde{Q}$, so the condition holds for $d\tilde{Q}$ sufficiently large.

As mentioned in the Introduction, this result contrasts with the literature on latent contracts (notably Parlour and Rajan (2001) and Attar et al. (2019a, 2019b)). Although our credit lines, never being drawn, resemble the latent contracts in that literature, the outcomes here do not resemble those there. With (i) exclusivity within periods, albeit arbitrarily short ones, and (ii) competition in bundles, not just loans, the non-competitive outcomes of that literature do not arise.

There is also a practical difference between credit lines in our model and latent (loan) contracts in the literature, namely that whereas a credit line is a contract agreed to between a borrower and a (potential) lender that must be honored, the latent contract is just an offer from a lender that can be retracted. That matters, because, in our model, lenders would prefer not to honor credit lines. At the time that they would be drawn, B is so indebted that the price $\tilde{p}$ is too high for the lender to break even. Whereas credit lines, which are always available per the contract, can support a variety of outcomes, the analogous latent contracts demand a stronger form of lender commitment.

5 Robustness and Extensions

We now relax several assumptions from the baseline model: We allow for (i) competition in loan-credit line bundles at all t , not just $t = 0$ (Section 5.1), (ii) revocation of the credit line by lenders with some probability (Section 5.3), (iii) uncertainty through shocks to the gains from trade between B and lenders (Section 5.2), and (iv) a reduced-form version of imperfect competition (Section 5.4). The proofs of the results are in the Supplemental Online Appendix.

5.1 Bundling at Any Date

In our baseline model, we assume that lenders compete by offering bundles of loans and credit lines at $t = 0$ and only loans thereafter, for $t > 0$. Here, we relax this

assumption and show that the outcome in Proposition 5 is robust to allowing competition in bundles for $t > 0$. To do so, we show that if B has debt Q_0^e and a credit line $(\tilde{p}, d\tilde{Q})$ as defined in Proposition 5 in place, there is no bundle $((p_t, dQ_t), (\tilde{p}_t, d\tilde{Q}_t))$ a lender can offer at $t > 0$ and make strictly positive expected profits. This is formalized in the following Proposition.

Proposition 6 (Bundling for $t > 0$). *The equilibrium allocations in Proposition 5 are robust to allowing lenders to compete in bundles at all t .*

The result says that the equilibrium in Proposition 5 survives competition in debt-credit line bundles for all t .

To establish the result we consider a fictitious lender that has, by fiat, the power to enforce exclusivity in the future. We show that even that lender does not have a profitable deviation and, therefore, a lender without that power does not either. The intuition is the same as for Proposition 4: The initial credit line deters subsequent ones, just as it does subsequent debt contracts in the baseline, as any increase in Q triggers the draw down of the original credit line, diluting the new debt so much that it is loss making.

5.2 Uncertainty

In our baseline model, we assume that there is no risk while the project is alive, only the termination date is random. We think that helps distinguish our theory from other theories of credit lines, which are generally based on insuring against adverse risks. Nonetheless, we relax that assumption here. Aside from showing robustness, we hint at how our mechanism generates something that looks like relationship banking.

We now assume that gains from trade can vary over time due to changes in the gains from trade/corporate tax: τ is a state variable.¹⁰ For a given credit line, the associated debt level is defined as in Definition 1. As in our baseline, we now consider

¹⁰So B's cost of debt $c_\tau = (1 - \tau)\delta$ is random; lenders' benefit δ is not.

a credit line $d\tilde{Q}$ with state-dependent price given by

$$\tilde{p}_\tau d\tilde{Q} + V_\tau(\mathcal{Q} + d\tilde{Q}) = V_\tau(\mathcal{Q}). \quad (26)$$

Note that the price is now contingent on the state, arguably capturing how credit line rates are conditioned on observable quantities, e.g., they are quoted in terms of spreads over the prevailing interest rate. We do not, however, need the size of the credit line $d\tilde{Q}$ to depend on the state, which also seems to accord with practice. As in Section 4, we maintain the assumption that credit lines can be offered only at $t = 0$ and that only loans are allowed thereafter.¹¹

To simplify the computation of the new value functions, we suppose that uncertainty is captured by a one-time shock to τ : With probability πdt , the corporate tax increases from τ_ℓ to τ_h and stays there forever. Let $\mathcal{Q}_\tau$ denote the optimum debt level conditional on the state, so

$$\tau\delta = \lambda'(\mathcal{Q}_\tau)\mathbb{E}[\tilde{V}_\tau(\mathcal{Q}_\tau)]. \quad (27)$$

In this scenario, the optimal strategy is to issue $dQ_0 = \mathcal{Q}_{\tau_\ell}$ and $dQ = \mathcal{Q}_{\tau_h} - \mathcal{Q}_{\tau_\ell}$ when the tax rate changes.

We show conditions under which that is indeed an equilibrium in Proposition 7. For τ_h , everything is as in the baseline as there is no uncertainty after the increase in the tax. Therefore, we have

$$\tilde{V}_{\tau_\ell} = \frac{y - c_{\tau_\ell}\mathcal{Q}_{\tau_\ell} + \pi\tilde{V}_{\tau_h}}{\rho + \pi + \lambda(\mathcal{Q}_{\tau_\ell})} \quad (28)$$

$$\tilde{V}_{\tau_h} = \frac{y - c_{\tau_h}\mathcal{Q}_{\tau_h}}{\rho + \lambda(\mathcal{Q}_{\tau_h})}. \quad (29)$$

The key twist relative to the baseline is what happens when τ changes, since, given B

¹¹That simplifies the proof here but an analog of Proposition 6, on the robustness of the equilibrium allocation to competition in bundles for $t > 0$, can be extended to this setting.

has outstanding debt, the market price will generally not serve to equate the marginal benefit and marginal cost in equation (27). Thus raising dQ_τ from the market could undermine the optimum. One straightforward way to avoid the problem is, in the spirit of Arrow–Debreu, to assume that debt issued at date 0 is contingent on the state. So the face value changes automatically from Q_{τ_ℓ} to Q_{τ_h} . We stick to that interpretation in our formal analysis. More realistically, this could be implemented as follows: B issues debt initially, and increases his debt when needed with his initial lender, perhaps by drawing partially on the line, with the line adjusting to deter new debt from the market. That would resemble the long-term/captive relationships we observe many borrowers have with a single lender.

The next result shows the robustness of our baseline analysis to uncertainty.

Proposition 7 (Uncertainty). *Let Q_τ be the optimal debt level with associated credit line $((\tilde{p}_\tau, d\tilde{Q}))_\tau$ given by equation (26). For $d\tilde{Q}$ sufficiently large, there is an equilibrium in which (i) the ratchet effect obtains (in that B draws if and only if $Q > Q_\tau$) and (ii) the exclusive equilibrium allocation is implemented.*

As in the baseline environment without uncertainty, the proof relies on the cost of another unit of debt decreasing as debt increases and on gains from trade being negative for high debt. With uncertainty, the debt level and the credit line price adjust to ensure that the analogous conditions hold state by state: When new gains from trade arise, the credit line is repriced after B issues the new debt with its pre-existing (original) lender.

5.3 Credit Line Revocation

In our baseline model, we assume that lenders fully commit to credit lines; they are never revoked. Here we relax that assumption. We assume that, conditional on being drawn, a lender honors the credit line with probability α and defaults, lending nothing, with complementary probability. One motivation for this is that the offering

lenders could be distressed themselves and thus unable to honor their commitments (cf. Chodorow-Reich and Falato (2022)).

As above, B draws the credit line $(\tilde{p}, d\tilde{Q})$ if and only if his debt is (strictly) above the associated $\mathcal{Q}$, defined, by analogy with Definition 1, as

$$\alpha(\tilde{p}d\tilde{Q} + V(\mathcal{Q} + d\tilde{Q})) + (1 - \alpha)V(\mathcal{Q}) = V(\mathcal{Q}). \quad (30)$$

Likewise, the condition for B not to want to borrow $dQ > 0$ (in which case he draws on $(\tilde{p}, d\tilde{Q})$) is analogous to inequality (20) with drawing the line replaced by drawing it with probability α :

$$pdQ + \alpha(\tilde{p}d\tilde{Q} + V(\mathcal{Q} + dQ + d\tilde{Q})) + (1 - \alpha)V(\mathcal{Q} + dQ) \leq V(\mathcal{Q}), \quad (31)$$

or, using B's indifference condition in equation (30) to rewrite the RHS and rearranging,

$$p \leq -\alpha \frac{V(\mathcal{Q} + dQ + d\tilde{Q}) - V(\mathcal{Q} + d\tilde{Q})}{dQ} - (1 - \alpha) \frac{V(\mathcal{Q} + dQ) - V(\mathcal{Q})}{dQ}. \quad (32)$$

Per the argument in Section 4.2 (cf. equation (21)), it is necessary and sufficient that the inequality holds for $dQ \rightarrow 0$, or that

$$p \leq -\alpha V'(\mathcal{Q} + d\tilde{Q}) - (1 - \alpha)V'(\mathcal{Q}). \quad (33)$$

From here, we have the next result:

Lemma 4 (No New Debt with Revocation). *Suppose B has debt $\mathcal{Q}$ and a revocable credit line $(\tilde{p}, d\tilde{Q})$ in place, such that B is indifferent to drawing the line at $\mathcal{Q}$.*

For $dt \rightarrow 0$, B prefers new debt (p, q) for some q to no loan if and only if inequality (33) is violated.

The result says that B does not want to borrow at a price below marginal cost. The

twist is that the marginal cost is conditional on drawing the line successfully.

We now turn to whether lenders are willing to lend. By the definition of Γ , a lender that anticipates that B will draw the credit line successfully with probability α is willing to offer a loan (p, dQ) if and only if

$$\alpha\Gamma(\mathcal{Q} + dQ + d\tilde{Q}) + (1 - \alpha)\Gamma(\mathcal{Q} + dQ) \geq p. \quad (34)$$

Just comparing inequalities (33) and (34) gives the next result:

Proposition 8 (Revocation). *Consider the setting of Proposition 4. B does not take on new debt at any price lenders will lend at if and only if*

$$\alpha\Gamma(\mathcal{Q} + d\tilde{Q}) + (1 - \alpha)\Gamma(\mathcal{Q}) \leq -\alpha V'(\mathcal{Q} + d\tilde{Q}) - (1 - \alpha)V'(\mathcal{Q}). \quad (35)$$

Re-writing (35) gives an expression for the debt levels that B can commit to with credit lines:

$$\Gamma(\mathcal{Q}) + V'(\mathcal{Q}) \leq \frac{\alpha}{1 - \alpha} \left(-\Gamma(\mathcal{Q} + d\tilde{Q}) - V'(\mathcal{Q} + d\tilde{Q}) \right), \quad (36)$$

where the term in parenthesis is positive for $d\tilde{Q}$ large by Assumption 2.

The analysis underscores the trade-off in the baseline model: You can support price above marginal cost $\mathcal{Q}$ to the extent that the price is below the marginal cost at $\mathcal{Q} + d\tilde{Q}$, except here the marginal cost conditional on drawing the line must be multiplied by the odds that the line is not revoked: $\alpha/(1 - \alpha)$. For α large, some commitment can be sustained; indeed, the full commitment outcome is attained in the baseline with $\alpha = 1$ (Proposition 5). However, as α gets small, B loses the ability to commit entirely, and the outcome approaches non-exclusive outcome $\Gamma(\mathcal{Q}) = -V'(\mathcal{Q})$ of Proposition 2.

5.4 Imperfect Competition

In our baseline model, we assume that there is perfect competition at each date. Here we describe how to capture imperfect competition in a minimal extension that mirrors our empirical proxies.

Suppose that, as in the baseline, B can get a credit line at date 0, but, unlike in the baseline, it has a fixed cost (κ). An instant later, at a date we might call $0+$, he enters the competitive market of the baseline with probability $1 - \mu$. Otherwise, he remains in a bilateral monopoly, corresponding to the exclusive benchmark of Proposition 1. Hence μ is a measure of search frictions or concentration. Since credit lines are redundant under exclusivity but implement commitment under non-exclusivity, B uses them when their value exceeds their cost, or

$$(1 - \mu)[V^e(0) - V(0)] \geq \kappa. \quad (37)$$

It follows immediately that B is more likely to have loans bundled with credit lines in more competitive markets, i.e. when μ is lower—credit lines matter only insofar as competition arises in the future—and when dilution costs are high, i.e. when $V^e(0) - V(0)$ is higher. This gives us a theoretical underpinning for our empirical use of market concentration as a proxy for dilution risk below.

6 Empirical Analysis

We now derive a set of empirical predictions from the theory and test them using data on loan originations, credit lines, and corporate balance sheets. We exploit two proxies for dilution risk and two sources of quasi-exogenous variation in the availability and credibility of credit lines to test the model’s key mechanisms. We then discuss alternative theories of credit lines and assess the extent to which they can account for the empirical patterns we document.

6.1 Empirical Tests

Here we summarize and test our main empirical predictions. We use data on loan originations from Dealscan and data on firm capital structure and characteristics from Capital IQ, Compustat, and CRSP. Our sample ranges from 2002 through 2019, unless otherwise stated. See Appendix A for details on the data and variable construction.

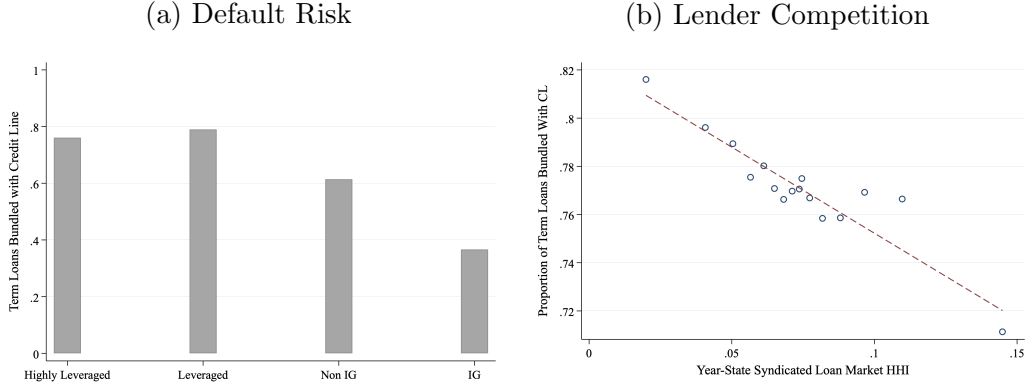
Empirically, we use two proxies for dilution risk. The first is default risk, since there is no dilution without default. The second is lender competition, since there is no dilution without competing lenders. We measure default risk using both Dealscan’s borrower market segment classification and firms’ cost of debt. We measure loan market competition with the Herfindahl-Hirschman Index (HHI) of the syndicated loan market for each state and year, calculated as the sum of squared market shares of all lenders in our sample. See Appendix A for more details.

Prediction 1. *Credit lines are bundled with loans, particularly for borrowers with high dilution risk.*

In the model, bundling a loan with a credit line allows lenders to offer more attractive terms by mitigating future dilution (Proposition 5). Thus bundling should be pervasive to the extent that dilution risk is.

- In our sample, 72% of term loans are bundled with a credit line at origination.
- Figure 1 (panel a) shows a positive relationship between bundling propensity and firm risk. Figure 1 (panel b) shows a negative relationship between bundling propensity and market concentration.

Figure 1: Bundling Propensity is Increasing in Firm Dilution Risk



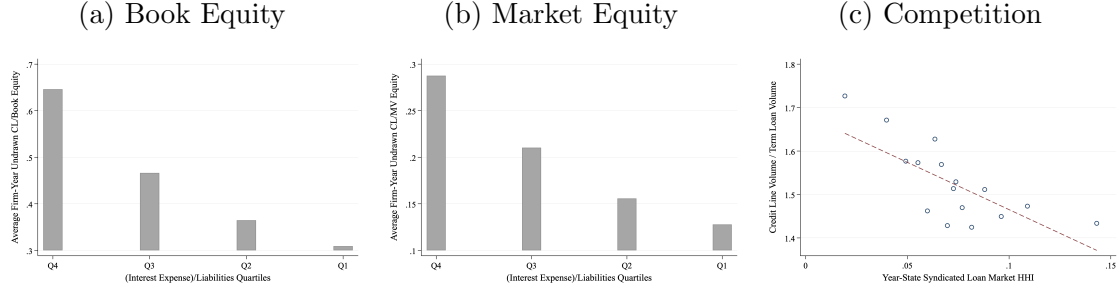
Panel (a) shows that a higher proportion of term loans are bundled with a credit line for riskier firms, where the safest borrowers are Investment Grade (“IG”), and the riskiest borrowers are “Highly Leveraged”. Panel (b) shows that bundling is more likely in competitive markets by plotting a bin scatter of the regression: $\text{bundling propensity}_{lst} = \alpha_{lt} + \beta \text{HHI}_{st} + \varepsilon_{lst}$, for lender l in state s in year t , which reveals a negative relationship between bundling and market concentration for a given lender. Here the outcome variable is the proportion of term loan deals that include a credit line among deals that lender l originates in state s in year t ; HHI_{st} is the syndicated loan market HHI in state s in year t ; and α_{lt} is a lender-year fixed effect. Data are from Dealscan.

Prediction 2. *Bundled credit lines are large, particularly for borrowers with high dilution risk.*

In the model, credit lines must be sufficiently large to deter future borrowing (Proposition 4). Intuitively, they must be large enough that drawing them eliminates the gains from additional debt issuance. This implies that credit lines should be large, and especially so in environments where dilution concerns are more severe.

- In our sample, on average, undrawn credit lines are 80% of book equity, 20% of market equity, and, at origination, 1.7 times the size of the term loan that they are bundled with (see the Supplemental Appendix).
- Figure 2 (panels a and b) shows a positive relationship between firm risk and credit line size. Figure 2 (panel c) shows a negative relationship between relative credit line size and market concentration.

Figure 2: Credit Line Size is Increasing in Firm Dilution Risk



Panels (a) and (b) show that undrawn CL are larger for riskier firms, as measured by quartiles of interest expense to total liabilities, where Q4 represents firms with the highest cost of debt. Panel (a) considers credit lines as a share of book equity and Panel (b) as a share of market equity, and data come from Capital IQ, Compustat, and CRSP. Panel (c) shows that credit lines are larger in more competitive markets by plotting a bin scatter of the regression $\left(\frac{\text{credit line volume}}{\text{term loan volume}}\right)_{lst} = \alpha_{lt} + \beta \text{HHI}_{st} + \varepsilon_{lst}$, for lender l in state s in year t using data from Dealscan. Here the outcome variable is the average volume of lender l 's credit lines relative to the bundled term loans in state s in year t ; HHI_{st} is the syndicated loan market HHI in state s in year t ; and α_{lt} is a lender-year fixed effect.

Prediction 3. *Credit lines deter subsequent borrowing, particularly for borrowers with high dilution risk.*

In the model, credit lines allow borrowers to commit not to dilute existing debt by deterring future borrowing (Proposition 5). This implies that weakening the availability or credibility of credit lines should increase subsequent borrowing, and especially so in environments where dilution concerns are more severe.

We test the prediction using two natural experiments. We first use quasi-random variation in credit line availability due to the Supplementary Leverage Ratio (SLR), which increased the cost of providing credit lines for some banks in 2018.

- First, we find that firms borrowing from affected banks are less likely to obtain credit lines. Consistent with the model, such firms are more likely to borrow again within a year (Table 1).

Table 1: Credit Lines Deter Subsequent Borrowing: At Origination

	(1)	(2)
	Bundling Propensity	Subsequent Borrowing
SLR $\times$ Post	−0.10*** (0.04)	0.07* (0.04)
SLR	0.02 (0.04)	−0.05 (0.05)
Post	0.01 (0.03)	−0.12*** (0.04)
Avg Lender Deals	−0.00 (0.00)	−0.00 (0.00)
Firm FE	Yes	Yes
Observations	4101	4101
Adjusted R^2	0.565	0.061

The table estimates for firm i 's deal d in year t : $Y_{idt} = \alpha_i + \beta \text{SLR}_{idt} \times \text{post}_t + \gamma \text{SLR}_{idt} + \delta \text{post}_t + \theta X_{idt} + \varepsilon_{idt}$. Here, Y_{idt} is an indicator for whether the term loan is bundled with a credit line in deal d (Column 1) or whether firm i borrows again within one year (Column 2); SLR_{idt} equals 1 if any lenders in the deal are subject to the SLR; post_t equals 1 if t is 2018 or later; X_{idt} includes the average number of deals that lenders participating in deal d engage in during year t to capture cross-sectional differences in lenders; α_i is a firm fixed effect. We restrict our sample to non-refinancing loan packages originated from 2010 to 2020 to focus on new borrowing decisions around the SLR shock. Standard errors are clustered at the year level and reported in parentheses. One, two, and three stars indicate statistical significance at the 10%, 5%, and 1% level, respectively.

We next test the same mechanism using quasi-exogenous variation in the likelihood that credit lines are revoked. Following Chodorow-Reich (2014) and Darmouni (2020), we proxy for revocation risk using negative shocks to lender health, measured separately for all lenders and for credit line lenders (see Appendix A).

- Consistent with the model, borrowers take on more debt following a negative shock to their credit line lenders, whereas a negative shock to all lenders reduces borrowing, as in the existing credit-supply literature (Table 2).

Table 2: Credit Lines Deter Subsequent Borrowing: Revocation

	(1)	(2)
	Borrowing	Leverage
Shock	-0.06** (0.03)	0.01 (0.03)
Shock CL	0.04*** (0.01)	0.02* (0.01)
Constant	0.09*** (0.02)	-0.03* (0.02)
Observations	3465	1800
Adjusted R^2	0.003	0.002

The table estimates for firm i : $\text{new debt}_i = \alpha + \beta \text{shock CL}_i + \gamma \text{shock}_i + \delta X_i + \varepsilon_i$, where the outcome variable is an indicator for borrowing in the syndicated loan market during the crisis period in Column (1), and the change in overall leverage from 2008 to 2009 in Column (2). Leverage is defined as the ratio of long-term debt to total assets, and balance sheet information comes from Compustat. The definition of crisis and normal periods are described in Appendix A. Robust standard errors are reported in parentheses. One, two, and three stars indicate statistical significance at the 10%, 5%, and 1% level, respectively.

- Table 3 shows that the effects of both shocks are stronger for riskier borrowers and in more competitive markets.

Table 3: Subsequent Borrowing and Dilution Risk

(a) SLR				
	Borrower Risk		Local Competition	
	(1) High	(2) Low	(3) High	(4) Low
SLR $\times$ Post	0.07** (0.02)	−0.00 (0.37)	0.09** (0.04)	0.01 (0.09)
SLR	−0.02 (0.05)	−0.21 (0.23)	−0.10 (0.13)	−0.05 (0.09)
Post	−0.13** (0.06)	0.02 (0.32)	−0.15 (0.08)	−0.10 (0.07)
Avg Lender Deals	−0.00 (0.00)	−0.00 (0.00)	0.00 (0.00)	−0.00 (0.00)
Firm FE	Yes	Yes	Yes	Yes
Observations	3818	175	1598	1409
Adjusted R^2	0.068	0.102	0.073	0.065

(b) Revocation				
	Borrower Risk		Local Competition	
	(1) High	(2) Low	(3) High	(4) Low
Shock	−0.05*** (0.02)	−0.17 (0.15)	−0.11*** (0.03)	−0.07** (0.03)
Shock CL	0.03*** (0.01)	0.03 (0.05)	0.04*** (0.01)	0.02** (0.01)
Constant	0.07*** (0.02)	0.24* (0.12)	0.12*** (0.03)	0.10*** (0.03)
Observations	5547	592	3102	2736
Adjusted R^2	0.003	−0.001	0.006	0.002

Panel A replicates the specification in Table 1, Column 2 across firm subsamples, while Panel B replicates Table 2, Column 1 across the same subsamples. Borrowers are classified as high risk if they fall into any of Dealscan’s non-investment-grade categories. Firms are considered to operate in highly competitive markets if the HHI of the state in which the loan is originated falls in the bottom half of the HHI distribution. Standard errors are reported in parentheses. One, two, and three stars indicate statistical significance at the 10%, 5%, and 1% level, respectively.

Prediction 4. *Credit lines are rarely drawn.*

In the model, credit lines serve as a commitment device. They are not drawn in our baseline equilibrium (Proposition 4), although they can be once uncertainty is introduced (Section 5). They could also be drawn for reasons outside our model, which we do not purport to rule out (Section 6.2). Nonetheless we study credit line utilization, which should be lower under our theory than others, and especially so in environments where dilution concerns are more severe.

- In our sample, 45% of borrowers utilize none of their credit line and the median utilization rate is 17%. Even in crisis times, when theories of credit lines as liquidity insurance suggest high utilization, the majority (54%) of the average credit line is not used. See the Supplemental Appendix for the distribution of credit line utilization in our sample overall, during the Great Financial Crisis, and at the onset of the covid-19 pandemic.

Prediction 5. *Rates on credit lines are relatively low ($\tilde{p}$ is relatively high).*

This follows from re-writing the indifference condition (19), recalling that V is convex, and using the condition for $\Gamma(Q^e + d\tilde{Q})$ in Proposition 4:

$$\tilde{p} = -\frac{V(Q^e + d\tilde{Q}) - V(Q^e)}{d\tilde{Q}} > -V'(Q^e + d\tilde{Q}) \geq \Gamma(Q^e + d\tilde{Q}). \quad (38)$$

This condition says that conditional on the line being drawn, the credit line is priced above the lender's value and would generate negative profits for the lender. This implies that credit line rates are relatively low compared to what standard pricing considerations for $d\tilde{Q}$ would suggest. Indeed, if they were drawn down they would not compensate lenders for the higher default risk that would arise.

- In our data, the median spread on credit lines (including utilization fees) is essentially the same as that on the associated term loan (see Table 4). This

Table 4: Rates on Credit Lines Are Relatively Low

	mean	p10	p25	p50	p75	p90
CL Rate - Term Loan Rate	-33.663	-122.195	-25.000	0.000	0.000	4.779
Observations	2870					

The table reports the rate spread for credit lines (when drawn) minus the rate spread for term loans within a package, (including the utilization fee) in basis points. Data are from Dealscan.

is striking because drawing a credit line would mechanically increase leverage. In our sample, drawdowns raise leverage by about 20%, corresponding to an absolute increase of roughly 0.04. Using Schwert’s (2020) estimate of the semi-elasticity of interest rates with respect to leverage (1.36), this increase in leverage would imply an increase in spreads of about 5.4 percentage points. In contrast, we observe no meaningful difference between credit line and term loan rates. This suggests that credit line pricing does not reflect the higher leverage associated with drawdowns, consistent with the model’s prediction that credit lines are priced as if they will remain undrawn.

6.2 Alternative Theories and Robustness

We now turn to two salient theories of credit lines in the literature and argue that some empirical findings that follow swiftly from our theory do not follow so naturally from them.

Liquidity insurance. One theory of credit lines is that they implement contingent liquidity insurance in environments with limited pledgeability (Holmström and Tirole (1998)). We think that is a compelling theory that complements, rather than compete with, our own. It is consistent with firms holding lines and drawing them in distress, something observed in the data. But there are several facts we uncover that this theory does not readily speak to: It does not inherently speak to (i) why credit lines

are systematically bundled with term loans at origination, (ii) why credit lines should be so large, (iii) why utilization remains low, even in crisis periods, (iv) why bundling propensity should increase with competition, or (v) why size should likewise.

We provide a direct test that distinguishes the liquidity-insurance view from our mechanism by examining how firms respond to shocks to credit line availability. Under a liquidity-insurance interpretation, firms that lose access to credit lines should increase borrowing primarily when they face liquidity shortfalls, and should use that borrowing to rebuild cash buffers. We test this prediction by relating subsequent borrowing to firms' pre-shock cash holdings and to changes in cash following the shock, in Table 5. We find that firms with higher pre-shock cash holdings are less likely to borrow overall, consistent with lower liquidity needs, yet conditional on exposure to credit-line shocks they are more likely to borrow again. Moreover, controlling for changes in cash does not attenuate the effect, which is inconsistent with a liquidity-hoarding interpretation in which firms borrow to rebuild cash buffers. These patterns are difficult to reconcile with a pure liquidity motive, and instead indicate that subsequent borrowing reflects substitution toward external debt rather than precautionary liquidity management.

Table 5: Credit Lines Deter Subsequent Borrowing: Revocation

	(1)	(2)
	Borrowing	Borrowing
Shock	-0.04 (0.04)	-0.04 (0.04)
Shock CL	0.00 (0.03)	0.00 (0.03)
Shock CL $\times$ Lag Cash/Assets	0.31* (0.16)	0.30* (0.16)
Lag Cash/Assets	-0.30*** (0.11)	-0.28*** (0.10)
Shock CL $\times$ Change in Cash		-0.01 (0.02)
Change in Cash		0.02 (0.02)
Constant	0.12*** (0.03)	0.12*** (0.03)
Observations	1844	1818
Adjusted R^2	0.001	0.003

The table replicates Table 2, Column 1 with additional control variables and interactions with Shock CL, the shock to credit line lenders. Column 1 includes the level and interaction of Lag Cash/Assets, firms' cash-to-assets ratio in 2008. Column 2 additionally includes the level and interaction of the log change in firm cash holdings from 2008 to 2009. Balance sheet information comes from Compustat. Robust standard errors are reported in parentheses. One, two, and three stars indicate statistical significance at the 10%, 5%, and 1% level, respectively.

Relationship lending. Another view is that credit lines are just part of an ongoing relationship between borrowers and lenders, facilitating repeated borrowing. This perspective can rationalize several features of the data, including why firms obtain multiple products from the same lender and therefore why they bundle credit lines with term loans. However, it does not provide as natural an explanation for several

other facts, such as why firms' overall borrowing and leverage should depend on credit line availability or why weakening credit line lenders' commitment leads to an increase in subsequent borrowing and leverage.

Moreover, the theory does not capture the cross-sectional patterns in our data. Because relationship lending is often associated with lenders' informational advantages or market power, we would expect the patterns predicted by the theory to be more pronounced when borrower-lender ties are strongest and/or outside options are limited. To the contrary, we find that bundling is more prevalent, credit lines are larger, and the increase in subsequent borrowing following shocks to credit-line credibility is stronger in more competitive markets.

7 Conclusion

We study a model that suggests that credit lines play a heretofore overlooked role. They can mitigate debt dilution. The theory suggests that the option to borrow—viz. a credit line—is valuable even if it is never exercised, explaining why credit lines are ubiquitous but rarely drawn. It also underscores how and why credit lines should be bundled with loans, a pervasive practice whose theoretical underpinnings have not been successfully identified.

Our paper contrasts with recent corporate finance papers on the leverage ratchet effect, suggesting that including credit lines in the contracting space makes the ratchet effect self-deterring, undoing its negative effects. It also contrasts with the literature on latent contracts, suggesting that the kinds of outcomes stressed in that literature might not obtain in dynamic environments under limited commitment.

A Details on Empirical Methodology

A.1 Data

For information on loan originations, we use US C&I syndicated loans from DealScan (2002–2019), defined as US-originated loans for “general purpose” or “working capital,” excluding loans to financials. We retrieve information on firm capital structure using Capital IQ annual filings, keeping the latest filing each year for US non-financial public firms matched to Compustat. Finally, we obtain firms’ yearly market value of equity by merging our Capital IQ-Compustat sample with monthly CRSP data and averaging market capitalization over each year. Our sample ranges from 2002 through 2019. Additionally, we use Capital IQ quarterly data in 2020Q1 to examine credit line usage at the onset of Covid-19.

A.2 Variable Construction

Loan market competition. We proxy local loan market competition using the Herfindahl-Hirschman Index (HHI) of the syndicated loan market for each state s and year t , calculated as the sum of squared market shares of all lenders in our sample. Following Chodorow-Reich (2014) and Darmouni (2020), we impute missing lender shares by computing average shares of lead arrangers and participants separately for each syndicate structure (defined by the number of lead arrangers and other participants) over the sample period among syndicates without missing data, and filling missing shares with the corresponding averages.

Borrower risk. We proxy for borrower risk using Dealscan’s market segment classifications, which range from safest (Investment Grade) to riskiest (Highly Leveraged).¹²

¹²Dealscan’s borrower market segment classifications depend on pricing thresholds as follows. Investment grade: Loan to borrowers rated BBB-/Baa3 or higher with pricing thresholds based on market trends which change over time. Non Investment grade: A loan which is not considered Investment Grade. Leveraged: Loan to borrowers rated BB+/Ba1 or lower with pricing thresholds based on market trends which change over time. Highly Leveraged: Leveraged loan with pricing 100 bps higher than the pricing thresholds set forth for a leveraged loan.

Lender health shocks. For each borrower, we construct overall lender health shocks following Chodorow-Reich (2014) and Darmouni (2020) and, analogously, for lenders with credit lines outstanding to the borrower. Specifically, we define $\Delta L_{b,-i}$ as the decrease in a bank b 's lending to firms $j \neq i$ in the crisis period vis-à-vis normal times:

$$\Delta L_{b,-i} := 1 - \frac{2 \sum_{j \neq i} L_{b,j,\text{crisis}}}{\sum_{j \neq i} L_{b,j,\text{normal}}}, \quad (39)$$

where $L_{b,j}$ is the effective number of loan facilities from b to j during normal and crisis times, defined as 10/2005–6/2007 and 10/2008–6/2009, respectively. The effective number of loan facilities is the number of loan facilities originated, with each weighted by the corresponding lender share. We again impute lenders' shares following Chodorow-Reich (2014) and Darmouni (2020), now calculating the average lender share of lead arrangers and participants separately for each syndicate structure during the time period surrounding the Global Financial Crisis, from 2004 through 2010. We exclude refinancings and amendments (except extensions) during crisis times, and restrict the sample to firms that borrowed during the normal period and banks that are present in both the normal and crisis period. We winsorize $\Delta L_{b,-i}$ at 2%. We then construct the shocks for each borrower i 's lenders and credit line lenders as weighted sums of $\Delta L_{b,-i}$ over lenders in a borrower i 's last pre-crisis syndicate:

$$\text{Shock}_i = \sum_{b \in S} \alpha_b \Delta L_{b,-i} \quad \text{Shock CL}_i = \sum_{b \in S} \alpha_b^{\text{CL}} \Delta L_{b,-i}, \quad (40)$$

where S is the set of lenders in borrower i 's last pre-crisis syndicate. For a lender $b \in S$, α_b is its average share across all loan facilities in the syndicate and α_b^{CL} is its share within the credit line facility. If borrower i 's last pre-crisis syndicate has no CL or a CL that matures prior to 2008, we set $\text{Shock CL}_i = 0$.

SLR deal. We classify a loan deal to be subject to the SLR if any of the banks that participate in the deal are subject to the SLR at the time when the loan is originated.

Cost of loan deal. We define the term loan rate as the all-in-spread-drawn, and the credit line rate as the all-in-spread-undrawn plus the utilization fee, averaging these rates across all term loans or credit lines within a deal, respectively.

B Proofs

Proof of Proposition 1

Here we apply a guess-and-verify approach: We assume that it is optimal to set $dQ_t = 0$ for all $t > 0$, solve for the optimal Q_0 , and then show that B cannot benefit by issuing again. Our problem is differentiable and well-defined, thus we can use the first-order approach for our Step 1 (see Assumption 1).

Step 1: Optimal issuance at date 0. The optimal date-0 issuance as if B never issues debt again, $Q_0 = dQ_0$, solves

$$\max_{Q_0} p_0 Q_0 + \frac{y - c Q_0}{\rho + \lambda(Q_0)} \quad (41)$$

$$\text{s.t. } p_0 \leq \frac{\delta}{\rho + \lambda(Q_0)}, \text{ with FOC}$$

$$p_0 + \frac{dp_0}{dQ_0} Q_0 - \frac{c + \lambda'(Q_0) \frac{y - c Q_0}{\rho + \lambda(Q_0)}}{\rho + \lambda(Q_0)} = 0, \quad (42)$$

Using lenders' binding participation constraint and computing the integrals (which is easy for $Q_t \equiv Q_0$), we can rearrange to find an expression for the optimal Q_0 given that $p_0 = \frac{\delta}{\rho + \lambda(Q_0)}$,

$$\tau \delta = \lambda'(Q_0) \Psi(Q_0). \quad (43)$$

Step 2: No issuance at date $t^+ > 0$. Now we verify that the marginal benefit from

having $dQ_{t^+} \neq 0$ for some $t^+ > 0$ is zero. The problem of choosing dQ_{t^+} is as follows:

$$\max_{dQ} p_0 Q_0 + \frac{y - c Q_0}{\rho + \lambda(Q_0)} + e^{-(\rho + \lambda(Q_0))t^+} \left(p_{t^+} dQ + \frac{y - c(Q_0 + dQ)}{\rho + \lambda(Q_0 + dQ)} - \frac{y - c Q_0}{\rho + \lambda(Q_0)} \right) \quad (44)$$

$$p_0 = \frac{\delta}{\rho + \lambda(Q_0)} + e^{-(\rho + \lambda(Q_0))t^+} \left(\frac{\delta}{\rho + \lambda(Q_0 + dQ)} - \frac{\delta}{\rho + \lambda(Q_0)} \right) \quad (45)$$

$$p_{t^+} = \frac{\delta}{\rho + \lambda(Q_0 + dQ)} \quad (46)$$

Plugging in the price functions, removing the terms where dQ does not show up, and combining the rest, it follows that $dQ_{t^+}^*$ must solve the following simplified problem

$$\max_{dQ} \frac{\delta}{\rho + \lambda(Q_0 + dQ)} (Q_0 + dQ) + \frac{y - c(Q_0 + dQ)}{\rho + \lambda(Q_0 + dQ)}. \quad (47)$$

Therefore,

$$\lambda'(Q_0 + dQ)\Psi(Q_0 + dQ) = \delta - c \equiv \tau\delta. \quad (48)$$

With this, equation (43) implies $dQ_{t^+}^* = 0$.

Proof of Proposition 2

The expression for V in equation (15) follows from substituting the equation from the optimal control (equation (14)) into the continuous-time HJB (equation (13)). The equation for p in (16), comes from differentiating the equation for V (equation (15)) and replacing V' with $-p$ from equation (14).

The issuance policy follows from the law of motion for the price,¹³

$$(\rho + \lambda(Q))p(Q) = \delta + p'(Q)q. \quad (49)$$

¹³That equation, which can be seen as the Black–Scholes differential equation for a derivative with price p written on an underlying Q following $dQ_t = q_t dt$, is the limit of the standard discounting formula $p(Q)(\rho + \lambda(Q)) = \delta dt + e^{-(\rho + \lambda(Q))dt} p(Q + dQ)$.

Using $p = -V'$, differentiate to get $p' = -V''$. Substituting both above and rearranging, gives the expression for q in the proposition. $\square$

Proof of Lemma 1

We show that $V' < 0$ and $V'' > 0$ by differentiation, starting from the expression for V in equation (15).

We have that

$$V'(Q) = -\frac{c + \lambda'(Q) V(Q)}{\rho + \lambda(Q)}, \quad (50)$$

which is negative.

We have that

$$V''(Q) = -\frac{\lambda''(Q) V(Q) + 2\lambda'(Q) V'(Q)}{\rho + \lambda(Q)}, \quad (51)$$

or, substituting back for V' and rearranging,

$$V''(Q) = \frac{1}{\rho + \lambda(Q)} \left(-\left(\lambda''(Q) - \frac{2\lambda'(Q)^2}{\rho + \lambda(Q)} \right) V(Q) + \frac{2\lambda'(Q) c}{\rho + \lambda(Q)} \right). \quad (52)$$

That is positive by Assumption 3. $\square$

Proof of Lemma 2

The argument is in the text. $\square$

Proof of Proposition 3

We show the result via the following lemmata.

- In Lemma B.1, we rule out smooth equilibria in which B does not draw for $Q > \bar{Q}$. That allows us to consider only drawing or jumping.
- In Lemma B.2, we rule out equilibria in which B does not draw today but draws later on. That allows us to consider either drawing immediately or never.

At that point, we have the corollary that, given our focus on Markov equilibria, in any candidate equilibrium in which B does not draw, there must be an absorbing state.

- In Lemma B.3, we show that for $Q > \mathcal{Q}$, B does not jump to any absorbing state $Q^+ > \mathcal{Q}$.
- In Lemma B.4, we show, symmetrically, that for $Q > \mathcal{Q}$, B does not jump to any absorbing date $Q^- < \mathcal{Q}$ under a condition on prices, which we show holds in Lemma B.5 and Lemma B.6.
- Finally, in Lemma B.7, we show that B does not draw for $Q < \mathcal{Q}$.

The proposition follows because, at that point, we have shown that (i) for $Q \geq \mathcal{Q}$, B either prefers to draw or to jump to absorbing state $\mathcal{Q}$ and B weakly prefers to draw at $\mathcal{Q}$ by the definition of the associated credit line, and (ii) for $Q < \mathcal{Q}$, B does not draw.

Lemma B.1. *Suppose $(\tilde{p}, d\tilde{Q})$ satisfies (18). Suppose also that the policy is smooth and the value function $\tilde{V}$ is differentiable. B prefers to draw for $Q > \mathcal{Q}$.*

Proof. Suppose (in anticipation of a contradiction) that B strictly prefers not to draw at Q . By continuity, there is a neighborhood of Q for which he strictly prefers not to draw. By smoothness, near Q , $\tilde{V}$ satisfies the HJB as if there were no credit line (i.e. $(\rho + \lambda(Q))\tilde{V}(Q) = y + p(Q)q - cQ + \tilde{V}'(Q)q$). That equation has the solution $(\rho + \lambda(Q))\tilde{V}(Q) = y - cQ$. Given V is convex, from (18) we get, $\tilde{V}(Q) \leq \tilde{p}d\tilde{Q} + V(Q + d\tilde{Q})$. That is the desired contradiction. $\square$

Lemma B.2. *Suppose $(\tilde{p}, d\tilde{Q})$ satisfies (18). If B prefers to draw at t^+ , he prefers to draw at $t < t^+$.*

Proof. Suppose (in anticipation of a contradiction) that B prefers to draw at $t^+ = t + dt$ but not at t . Keeping in mind that, by Lemma B.1, it suffices to focus on jumps

and denoting Q_t by Q and Q_{t^+} by Q^+ , that says:

$$\tilde{p}d\tilde{Q} + V(Q + d\tilde{Q}) < \tilde{V}(Q) \leq p^+(Q^+ - Q) + \tilde{p}d\tilde{Q} + V(Q^+ + d\tilde{Q}), \quad (53)$$

where p is the issuance price and the discount rate $\rho + \lambda(Q)$ vanishes because $e^{-(\rho + \lambda(Q))dt} \rightarrow 1$ as $dt \rightarrow 0$.

Since B draws immediately at t^+ , we know that $p = -V'$ by Proposition 2, substituting and rearranging, we have that

$$V(Q^+ + d\tilde{Q}) - V(Q + d\tilde{Q}) > V'(Q^+ + d\tilde{Q})(Q^+ - Q). \quad (54)$$

Which is false for $Q = Q^+$ and contradicts the convexity of V for $Q \neq Q^+$. $\square$

The analysis so far says that it suffices to compare drawing immediately to a sequence of jumps $\{Q_{t_n}\}_n$. But, by the Markov assumption, at each Q_{t_n} , B either jumps immediately or stays there forever. (Staying at Q_{t_n} for an interval and then jumping later would violate the Markov assumption, which implies that at Q_{t_n} you either jump or not.) Given our restriction to non-explosive processes, immediate jumps are telescoping—they cancel out. So it is sufficient to compare drawing to jumping to an absorbing state. That is what we do in the next two lemmata.

Lemma B.3. *There is no absorbing state with $Q^+ > Q$.*

Proof. Suppose (in anticipation of a contradiction) that Q^+ is an absorbing state. B must prefer not to draw at Q^+ :

$$V(Q^+) \geq \tilde{p}d\tilde{Q} + V(Q^+ + d\tilde{Q}), \quad (55)$$

having used that, per Proposition 2, $V(Q^+)$ is both the value function absent a CL and the value in an absorbing state Q^+ (with or without CL). Combined with condition

(19), that $\tilde{p}d\tilde{Q} = V(\mathcal{Q}) - V(\mathcal{Q} + d\tilde{Q})$, we have

$$V(\mathcal{Q} + d\tilde{Q}) - V(\mathcal{Q}) \geq V(Q^+ + d\tilde{Q}) - V(Q^+), \quad (56)$$

which, given $Q^+ > \mathcal{Q}$, contradicts the convexity of V . $\square$

Lemma B.4. *Suppose $\mathcal{P} < \delta/(\rho + \lambda(Q^-))$ for $Q^- < \mathcal{Q}$, where $\mathcal{P}$ is the price at $\mathcal{Q}$. For $Q > \mathcal{Q}$, B prefers to jump to $\mathcal{Q}$ than to any possible absorbing state $Q^- < \mathcal{Q}$.*

Proof. We want to show that

$$\mathcal{P}(\mathcal{Q} - Q) + \tilde{V}(\mathcal{Q}) \geq p^-(Q^- - Q) + V(Q^-). \quad (57)$$

As B always has the option not to draw, his payoff from keeping the option $\tilde{V}$ must be at least high as that from exercising it:

$$\tilde{V}(\mathcal{Q}) \geq \tilde{p}d\tilde{Q} + V(\mathcal{Q} + d\tilde{Q}) = V(\mathcal{Q}), \quad (58)$$

where the last equality is condition (18). Thus, it suffices to show that inequality (57) holds for $V(\mathcal{Q})$ in place of $\tilde{V}(\mathcal{Q})$:

$$\mathcal{P}(\mathcal{Q} - Q) + V(\mathcal{Q}) \geq p^-(Q^- - Q) + V(Q^-). \quad (59)$$

Now given that $Q - \mathcal{Q} > 0$, if $p^- > \mathcal{P}$, it suffices to show the inequality with $\mathcal{P}$ replaced with p^- :

$$p^-(Q - Q^-) - p^-(Q - \mathcal{Q}) \geq -(V(\mathcal{Q}) - V(Q^-)) \quad (60)$$

or, given $\mathcal{Q} - Q^- > 0$,

$$p^- \geq -\frac{V(\mathcal{Q}) - V(Q^-)}{\mathcal{Q} - Q^-}. \quad (61)$$

Recall that Q^- is a supposed absorbing state, so $p^- = \gamma(Q^-)/(\rho + \lambda(Q^-))$. The

inequality becomes

$$\frac{\delta}{\rho + \lambda(Q^-)} \geq -\frac{V(Q) - V(Q^-)}{Q - Q^-}, \quad (62)$$

a sufficient condition for which is

$$\frac{\delta}{\rho + \lambda(Q^-)} \geq -V'(Q^-), \quad (63)$$

or, from Proposition 1,

$$\tau\delta \geq \lambda'(Q^-)V(Q^-) \quad (64)$$

which is satisfied for Q^- below Q^e (cf. condition (8)). $\square$

Lemma B.5. *Suppose that Q^0 and Q^1 are both absorbing states with $Q^0 < Q^1 < Q^e$. B prefers to jump from $Q > Q^1$ to Q^1 than to Q^0 .*

Proof. We want to show that

$$p^1(Q^1 - Q) + V(Q^1) > p^0(Q^0 - Q) + V(Q^0) \quad (65)$$

or, rearranging,

$$-\left(V(Q^1) - V(Q^0)\right) < p^0(Q - Q^0) - p^1(Q - Q^1) \quad (66)$$

$$= p^0(Q - Q^0) - \underbrace{p^0(Q - Q^1) + p^0(Q - Q^1)}_{=0} - p^1(Q - Q^1) \quad (67)$$

$$= p^0(Q^1 - Q^0) + (p^0 - p^1)(Q - Q^1). \quad (68)$$

That holds whenever

$$-\frac{V(Q^1) - V(Q^0)}{Q^1 - Q^0} < p^0 + (p^0 - p^1)\frac{Q - Q^1}{Q^1 - Q^0}. \quad (69)$$

Given V is convex, $p^0 = \delta/(\rho + \lambda(Q_0))$, and the last term is positive ($p^0 > p^1$, $Q > Q^1$,

and $Q^1 > Q^0$), a sufficient condition is that

$$-V'(Q^0) < \frac{\delta}{\rho + \lambda(Q^0)}, \quad (70)$$

which is satisfied since $Q^0 < Q^e$. $\square$

Lemma B.6. *For $Q^- < \mathcal{Q}$, $p^- > \mathcal{P}$.*

Proof. Suppose (in anticipation of a contradiction) that $p^- < \mathcal{P}$. Given the arguments above, in particular the “corollary” argument following Lemma B.2, we know three things:

1. Q_t must jump down from $\mathcal{Q}$ (otherwise the price would be higher);
2. Q_t must jump to an absorbing state immediately, say Q^b (otherwise the policy would either be smooth, violating Lemma B.1, or would stay at a point for some time before jumping, violating the Markov property);
3. the absorbing state must be below Q^- , $Q^b < Q^-$ (otherwise the price at Q^b would be above that at Q^- , violating the hypothesis that $\mathcal{P} > p^-$).

B’s policy must, of course, be optimal: At $\mathcal{Q}$, he must prefer to jump to Q^b than to Q^- . But that contradicts Lemma B.5. We conclude that $p^- > \mathcal{P}$, as desired. $\square$

Lemma B.7. *For $Q < \mathcal{Q}$, B prefers not to draw.*

Proof. Here we show that when $Q < \mathcal{Q}$, B prefers to jump to $\mathcal{Q}$ than to draw the credit line (which implies only that drawing the line is not optimal, not that jumping to $\mathcal{Q}$ is):

$$\tilde{p}d\tilde{Q} + V(Q + d\tilde{Q}) < \mathcal{P}(\mathcal{Q} - Q) + \tilde{V}(\mathcal{Q}). \quad (71)$$

Using that, by the definition of the associated debt level $\mathcal{Q}$, $\tilde{V}(\mathcal{Q}) = \tilde{p}d\tilde{Q} + V(\mathcal{Q} + d\tilde{Q})$, inequality (71) becomes

$$V(Q + d\tilde{Q}) < \mathcal{P}(\mathcal{Q} - Q) + V(\mathcal{Q} + d\tilde{Q}) \quad (72)$$

or

$$\mathcal{P} > -\frac{V(\mathcal{Q} + d\tilde{Q}) - V(Q + d\tilde{Q})}{\mathcal{Q} - Q}. \quad (73)$$

By the convexity of V , it suffices that $\mathcal{P} > -V'(\mathcal{Q} + d\tilde{Q})$. That is the price if B draws at $\mathcal{Q}$ from Proposition 2. Since $Q < \mathcal{Q}$, that price is less than the price if B draws at Q and, since $\lambda' > 0$, it is also less than the price if $\mathcal{Q}$ is an absorbing state. Given we have established that drawing at Q and jumping to an absorbing state are the only two possibilities, the inequality is satisfied. $\square$

Proof of Lemma 3

Existence (of $\tilde{p} > 0$) follows immediately from V being decreasing: Just set

$$\tilde{p} := -\frac{V(Q + d\tilde{Q}) - V(Q)}{d\tilde{Q}}. \quad (74)$$

Uniqueness follows immediately from V being convex: $(V(Q + d\tilde{Q}) - V(Q))/d\tilde{Q}$ is strictly monotonic, hence intersects the constant $\tilde{p}$ at most once.

Proof of Proposition 4

We prove the proposition in three steps. First, we show that with a credit line satisfying the conditions of the proposition in place, then $\mathcal{Q}$ is an absorbing state (Lemma B.8). Second, we use this result to characterize the value function when such a credit line is in place (Lemma B.9). Finally, we verify that such a credit line always exists (Lemma B.10).

Lemma B.8. *If a credit line $(\tilde{p}, d\tilde{Q})$ satisfying the conditions of the Proposition is in place, then $\mathcal{Q}$ is an absorbing state.*

Proof. We show first that there is no issuance, i.e. no lender would buy at a price at which B would sell:

- Proposition 3 and the tie-breaking rule in Assumption 4 imply that lenders know B will draw for $Q > \bar{Q}$. Now the result follows from Lemma 2 and inequalities (21) and (23) as follows: For $Q_t > \bar{Q}$, $\Gamma(Q_t) = \frac{\delta}{\rho + \lambda(Q_t)}$ with $\lambda(Q) \rightarrow \infty$ as $Q \rightarrow \infty$. Thus, as condition (24) in the statement is satisfied, lenders do not lend at any price B would accept.

We now show that there are no buy backs at $\bar{Q}$, i.e. no lender would sell at a price B would buy:

- As $\bar{Q} \leq Q^e$, it follows from Assumption 2 that there are also no gains from offering debt buybacks.

We conclude that $\bar{Q}$ must be an absorbing state—there is no issuance and no buyback—and so $\tilde{V}(\bar{Q}) = V(\bar{Q})$. □

Lemma B.9. *If $\bar{Q}$ is an absorbing state, then the value function is as stated in Proposition 4.*

Proof. The result follows from what we have established already:

1. For $Q > \bar{Q}$, B draws the credit line by Proposition 3 and Assumption 4.
2. For $Q < \bar{Q}$, B jumps to absorbing state $\bar{Q}$ by Proposition 3 (see Lemma B.7 in the proof).
3. $\bar{Q}$ is an absorbing state by Lemma B.8 above. □

Lemma B.10. *Given $\tilde{V}$ above, a credit line satisfying the properties in the Proposition exists.*

Proof. In light of Lemma B.8 and Lemma B.9 above, the result follows immediately from Lemma 3. □

Proof of Proposition 5

First, suppose (in anticipation of a contradiction) that B takes up any bundle inducing a different outcome at date 0 in equilibrium. By Proposition 4, another lender can offer a bundle that implements Q^e and, by Proposition 1, it breaks even at p^e . By the definition of the full-commitment outcome, B is strictly better off. Thus, since B accepts at most one contract at each date, there is $\epsilon > 0$ such that the lender can offer $(p^e - \epsilon, Q^e)$ and both B and the lender (that was previously getting zero) are strictly better off. That is a profitable deviation and therefore a contradiction to the proposed equilibrium. Second, B is indifferent between drawing or not at all time, so he does not draw. □

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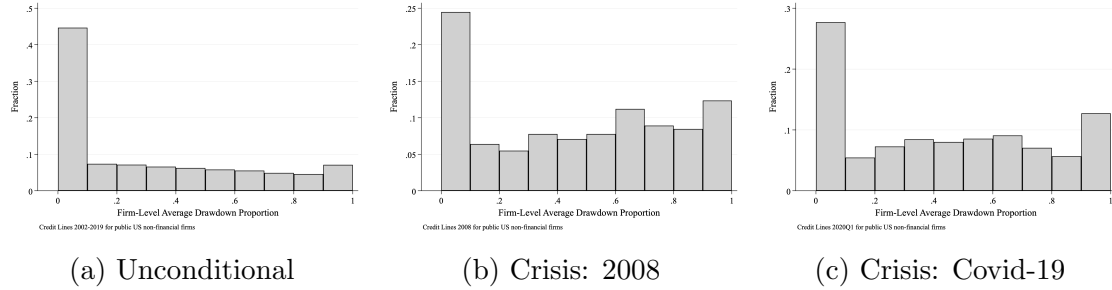
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SUPPLEMENTAL ONLINE APPENDIX

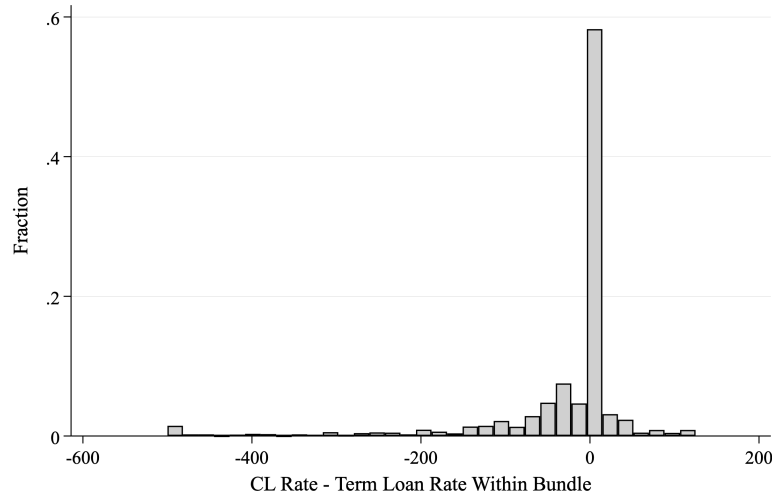
C Additional Figures and Tables

Figure 3: Credit Lines are Rarely Drawn



Panel (a) shows the annual utilization of credit lines from 2002 through 2019, panel (b) shows the annual utilization of credit lines during 2008, panel (c) shows the quarterly utilization of credit lines during 2020 Q1. Data are from Capital IQ and Compustat, covering US public non-financial firms.

Figure 4: Rates on Credit Lines are Relatively Low



This figure plots the rate spread for credit lines (when drawn) minus the rate spread for term loans within a package, in basis points. Data are at the loan package level from Dealscan from 2002 through 2019, covering US C&I syndicated loans and excluding financials. Differences are winsorized at 1%.

Table 6: Credit Lines Are Large

	Mean	p1	p10	p25	p50	p75	p90	p99
Credit Line / Term Loan	1.725	0.029	0.088	0.167	0.480	1.519	4.000	18.301
Undrawn CL /Book Equity	0.808	0.000	0.032	0.110	0.233	0.438	0.814	4.872
Undrawn CL /Market Equity	0.208	0.000	0.015	0.046	0.106	0.223	0.418	1.641

The table reports various measures of the size of firms’ credit lines. Data for the first row are from Dealscan, and for the remainder of the table are from our Capital IQ, Compustat, and CRSP.

Table 7: Credit Lines are Rarely Drawn

	mean	p10	p25	p50	p75	p90
2002–2019	0.296	0.000	0.000	0.170	0.548	0.836
2008 GFC	0.460	0.000	0.104	0.474	0.752	0.927
2021Q1 Covid	0.425	0.000	0.045	0.416	0.707	0.953

The table reports the distribution of credit line utilization unconditionally (2002 through 2019) and during crisis times (2008 and 2020Q1). Data are from Capital IQ and Compustat covering US public non-financial firms, and are annual at the firm level for the first two rows and quarterly at the firm level for the Covid-19 period.

D SLR Institutional Details

The SLR was introduced as part of the US implementation of the Basel III framework. The SLR was adopted in September 2014, and became effective on January 1st, 2018. It is a non risk-based leverage ratio that depends on the banks’ total leverage exposure, which includes a bank’s on and off-balance sheet exposures. Banks with greater than \$250B in total assets must maintain a tier 1 leverage ratio that is above a certain threshold (for instance, for GSIBs this threshold is 5%).

$$\text{SLR} \equiv \frac{\text{Tier 1 Capital}}{\text{Total leverage exposure}} \quad (75)$$

The SLR is considered to be the binding leverage ratio for these banks; undrawn credit lines, which are off-balance sheet, contribute to the denominator of the ratio.

E Proofs of Section 5

Proof of Proposition 6

We show the result via the following lemmata.

- In Lemma E.1, we establish that if a fictitious lender that, by fiat, has the power to enforce exclusivity in the future cannot make positive expected profits, neither can one without that power. We denote the fictitious lender by L^* .
- In Lemma E.2, we show that such a lender L^* indeed cannot profit by offering any debt-credit line bundle.

Lemma E.1. *Consider a (fictitious) lender L^* that can prevent all new lenders from entering in the future. (L^* has no power to prevent B from drawing existing credit lines or to prevent some lenders and not others from entering. The contract space is the same and there is no re-contracting.)*

If L^ cannot make positive profits, no other (real) lender can either.*

Proof. By construction, when L^* offers a contract it has a larger feasible set than any other lender. Therefore its profit must be weakly higher. The result follows. $\square$

Lemma E.2. *Define the lender L^* as in Lemma E.1.*

If B takes new debt from L^ , he draws his existing credit line.*

Proof. There are two cases: L^* enforces exclusivity or does not.

- Case 1: L^* enforces exclusivity. First observe that, by Proposition 2, B 's continuation value V under non-exclusivity with no credit lines coincides with that if B never borrows again.¹⁴ That says that his continuation value conditional on drawing the credit lines is V , as in the baseline, so the ratchet effect applies (Proposition 3): If B takes debt from L^* , he will draw his credit line $(\tilde{p}, d\tilde{Q})$.

¹⁴If B never borrows again and maintains debt Q forever, he gets $\int_0^\infty e^{-\rho t} (y - c(Q)) dt = (y - c(Q))/\rho$, which is the value $V(Q)$ in Proposition 2.

- Case 2: L^* does not enforce exclusivity. Here the argument is the same: Per the baseline, B draws the existing line (Proposition 3).

The result now follows from the anti-ratchet effect (Proposition 4): Anticipating that B will draw his original credit line, there is no price at which L^* will lend. $\square$

Proof of Lemma 4. The result follows from substituting $V' = -(c + \lambda'V)/(\lambda + \rho)$ from equation (16) into equation (33). $\square$

Proof of Proposition 7

We take a guess-and-verify proof. We guess that the value function with the credit line $((\tilde{p}_s, d\tilde{Q}))_s$ associated with debt levels $(\mathcal{Q}_s)_s$ is follows:

$$\tilde{V}_s(Q) = \begin{cases} \Gamma_s(\mathcal{Q}_s - Q) + V_s(\mathcal{Q}_s) & \text{if } Q \leq \mathcal{Q}_s, \\ \tilde{p}_s d\tilde{Q} + V_s(Q + d\tilde{Q}) & \text{otherwise,} \end{cases} \quad (76)$$

where Γ_s is the equilibrium price in state s .

Part (i): Ratchet effect. We first show that B draws if $Q > \mathcal{Q}_s$ and then that he does not if $Q \leq \mathcal{Q}_s$.

- *B draws if $Q > \mathcal{Q}_s$.* B prefers to draw if the value from doing so exceeds that from not:

$$\tilde{p}_s d\tilde{Q} + V_s(Q + q + d\tilde{Q}) \leq \tilde{V}_s(Q + q) \quad (77)$$

$$= \tilde{p}_s d\tilde{Q} + V_s(Q + q + d\tilde{Q}) \quad (78)$$

having substituted from our guess for the value function (equation 76) in the last step. The above says that B is always indifferent, implying optimality.

- *B does not draw if $Q < \mathcal{Q}_s$.* It suffices to show that B prefers not to draw and then immediately to issue $\mathcal{Q}_s - Q$ at price Γ_s than to draw (and subsequently to follow the optimal strategy):

$$\Gamma_s(\mathcal{Q}_s - Q) + \tilde{V}(Q + (\mathcal{Q}_s - Q)) \geq \tilde{p}d\tilde{Q} + V(Q + d\tilde{Q}) \quad (79)$$

or, using the definition of the associated credit line in equation (26) to substitute for $\tilde{p}d\tilde{Q}$, that

$$\Gamma_s(\mathcal{Q}_s - Q) + \tilde{V}(\mathcal{Q}_s) \geq -V_s(\mathcal{Q}_s + d\tilde{Q}) + \tilde{V}_s(\mathcal{Q}_s) + V(Q + d\tilde{Q}). \quad (80)$$

Given $\mathcal{Q}_s > Q$ in this case, that says that

$$\Gamma_s \geq -\frac{V_s(\mathcal{Q}_s + d\tilde{Q}) - V_s(Q + d\tilde{Q})}{\mathcal{Q}_s - Q}. \quad (81)$$

Since V_s is convex and $d\tilde{Q} > 0$, it suffices to show that

$$\Gamma_s \geq -\frac{V_s(\mathcal{Q}_s) - V_s(Q)}{\mathcal{Q}_s - Q}, \quad (82)$$

which holds by the definition of $\mathcal{Q}_s$ as the social optimum—the gains from issuance $\Gamma_s(\mathcal{Q}_s - Q)$ exceed the decrease in the value function $V_s(Q) - V_s(\mathcal{Q}_s)$.

Part (ii): Optimality. The result that the optimum can be implemented follows from Proposition 4 and Proposition 5. $\square$

Proof of Proposition 8

Immediate from inequalities (33) and (34). $\square$